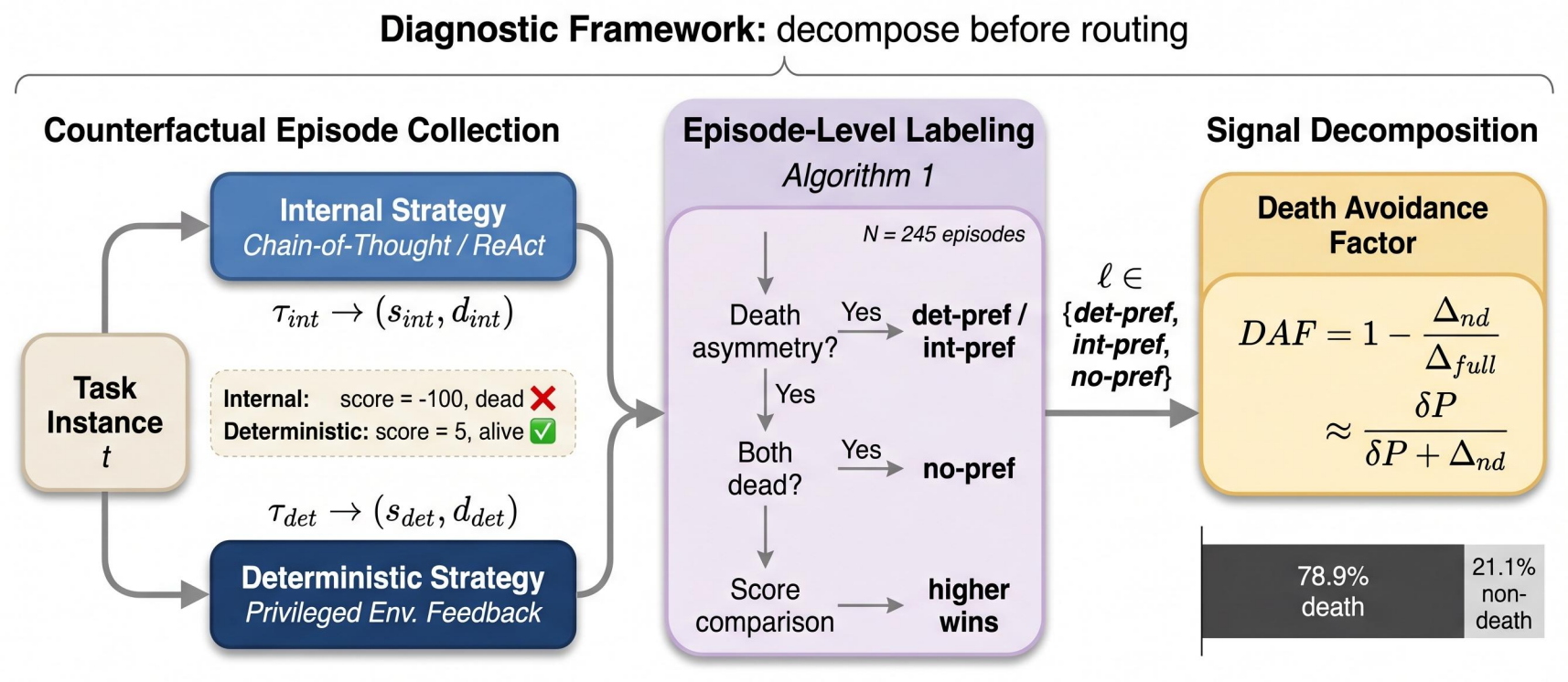

Diagnostic Framework: decompose before routing
Counterfactual Episode Collection
Task Instance $t$
Internal Strategy
Chain-of-Thought / ReAct
$\tau_{int} \rightarrow (s_{int}, d_{int})$
Internal: score = -100, dead
Deterministic: score = 5, alive
$\tau_{det} \rightarrow (s_{det}, d_{det})$
Deterministic Strategy
Privileged Env. Feedback
Episode-Level Labeling
Algorithm 1
N = 245 episodes
Death asymmetry?
Yes
det-pref / int-pref
Yes
Both dead?
Yes
no-pref
Score comparison
higher wins
$\ell \in$ {det-pref, int-pref, no-pref}
Signal Decomposition
Death Avoidance Factor
$DAF = 1 - \frac{\Delta_{nd}}{\Delta_{full}}$
$\approx \frac{\delta P}{\delta P + \Delta_{nd}}$
78.9% death
21.1% non-death